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Product Category Rule (PCR) Guidance for Building-Related Products and Services

Part B: Water Cooled Chiller EPD Requirements

UL 10010-4

www.ul.com/businesses/environment



Product Category Rules Part B: Water Cooled Chiller EPD Requirements, UL 10010–4

Second Edition, Dated June 29, 2018

Summary of Topics

Version 2.0 of the Product Category Rules Part B: Water Cooled Chiller EPD Requirements, UL 10010–4 has been issued.

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Publisher:

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Tracking of versions

Version	Comments	History
1.0	Preliminary version 1.0 by IBU – Product Category Rules Water Chilled Coolers confirmed by the Advisory Board (SVA)	May 2011
1.1	NA Addendum by UL Environment	January 2015
2.0	Update and revision to conform with ISO 21930: 2017, USGBC Guidance and UL Part A/Part B Structure	June, 2018

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Editor's Note:

This PCR is based on revisions made to the Institute of Construction and Environment (Institut Bauen und Umwelt e.V., or IBU) standard Part B PCR structure.

The revisions are not modifications to the overall methodology or structure of the IBU PCR, but are intended to reflect practices, methods and requirements that are specific to North America.

This PCR is valid for a period of five (5) years, set to expire in June, 2022.

I. Background Information and Acknowledgements

This PCR was developed to address the product group specific rules for the creation of Environmental Product Declarations (EPD) for water cooled chillers.

Other PCRs considered in the original development of this PCR include:

- Water Cooled Chiller PCR confirmed by the SVA, the Institute of Construction and Environment (Institut Bauen und Umwelt e.V., or IBU), Preliminary Draft v 1.0, May 2011¹

This PCR assumes a 75 year building service life to be consistent with ASTM E2921 (2013, Section 6.1.1), and ASHRAE 189.1 (2014, Section 9.5.1).

Interested Parties

This Part B has been prepared with input from the following stakeholders:

¹ At the time of publication, this PCR document was a preliminary version given EN 15804:2012 had yet to be published.



Trade Associations

- Air-Conditioning, Heating, and Refrigeration Institute (AHRI)

Manufacturers/Consultants

- Carrier
- Daikin
- Dimplex
- Trane

Governance

Trane sponsored the update of this PCR and is a member of the Air-Conditioning, Heating, and Refrigeration Institute (AHRI), the major industry accounting for more than 90% of HVACR and water heating residential and commercial equipment manufactured and sold in North America. AHRI's role is to inform its members of important industry developments and to represent their interests in projects such as the update of a PCR affecting their products. This is important because it effectively demonstrates that a large percentage of the chillers industry is represented in the effort to renew the PCR for water chiller products.

Involvement of Interested Parties

AHRI informed their members of the water chiller PCR update through their regularly scheduled association and association committee meetings, newsletters, e-mail messages, and similar types of outreach. Trade associations operate at the behest of its members, and the fact that trade associations are participating in the update of a PCR for chiller products is an indication that their memberships are aware of this project and have authorized their association to represent them in this important endeavour.

Per the USGBC PCR Guidance, 90% of the industry as represented by regional market volume was included in the update to this PCR with a minimum of three companies.

UL Environment posted an open call for participation in this PCR update in September 2017 via its standards website, social media outlets, and outreach to original committee stakeholders and LCA practitioners. UL Environment also pursued direct outreach with manufacturers outside of AHRI in the markets represented in this PCR.

Update Process

The PCR shall be revised five (5) years from the publication date. The PCR shall be revised before the five year date if the following occurs in the industry:

- The industry desires an update
- Core governing standards ISO 14040, 14044, 14025, 21930 are updated with substantial material changes

Review

This Part B was reviewed by the following:

Lise Laurin
EarthShift Global, LLC
lise@earthshiftglobal.com

Sean Beilman
BCER Engineering, Inc.
sbeilman@bcer.com

François Charron-Doucet
Group AGÉCO
Francois.charron@groupeageco.ca

II. Scope

This document contains the Product Category Rule (PCR) requirements for Water Cooled Chiller Environmental Product Declarations (EPDs) published in coordination with the EN 15804 and ISO 21930 standards. The requirements for the background Life Cycle Assessment (LCA) project report used to inform the EPD are contained in UL Environment's Part A: Life Cycle Assessment Calculation Rules and



Report Requirements. This Part B document, coupled with the Part A, conforms to the EN 15804, ISO 21930, ISO 14025, and the US Green Building Council PCR Guidance sustainability standards for EPD reporting.

This PCR does not address requirements for creating an Industry Wide EPD given the industry does not anticipate their creation.

General Guidance

The scope of this PCR applies to the product group water cooled chillers (referred to as 'chillers') that are used for the purpose of cooling water in an air conditioning system, or process cooling system of a commercial building or building complex. Chillers includes all commercially available chiller products according to the standards or technical approvals shown under Section 9.

This PCR applies to the entirety of a packaged product intended for individual sale.

Applicable Products

The following CSI code covers the scope of this Part B:

- 15620 (pre-2016)
- 23 64 00 Packaged Water Chillers:
 - o 23 64 13 Absorption Water Chillers
 - o 23 64 13.13 Direct-Fired Absorption Water Chillers
 - o 23 64 13.16 Indirect-Fired Absorption Water Chillers
 - o 23 64 16 Centrifugal Water Chillers
 - o 23 64 16.16 Water-Cooled Centrifugal Water Chillers
 - o 23 64 19 Reciprocating Water Chillers
 - o 23 64 23 Scroll Water Chillers
 - o 23 64 23.16 Water-Cooled Scroll Water Chillers
 - o 23 64 26 Rotary-Screw Water Chillers
 - o 23 64 26.16 Water-Cooled, Rotary-Screw Water Chillers
 - o 23 64 33 Modular Water Chillers
 - o 23 64 33.16 Water-Cooled, Modular Water Chillers

Non-Applicable Products

Chillers that are air-cooled are not included :

- 23 64 23.13 Air-Cooled Scroll Water Chillers
- 23 64 26.13 Air-Cooled, Rotary-Screw Water Chillers
- 23 64 33.13 Air-Cooled, Modular Water Chillers

System Boundary

The system boundary for EPDs created using this PCR is cradle to grave.

The EPD requirements include:

- Requirements of the EN 15804 standard as a European core EPD for the purpose of consistency between declarations in Europe and the United States.
- Requirements of the ISO 21930:2017 standard for the purpose of global consistency

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- The calculation rules for the Life Cycle Assessment and Requirements on the Project Report are specified in “Part A: Calculation Rules for the Life Cycle Assessment and Requirements on the Project Report,” available at <http://www.shopulstandards.com>

III. Industry Average EPD Requirements

Given the specificity of the products within the water cooled chiller category, rules for the industry average EPD could not be developed.

IV. Product-specific LCA calculation rules

Product-specific LCA calculation documents exist in accompanying templates to this PCR:

Centrifugal Chillers Average Efficiency Calculator.xlsx

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1. General Information

PROGRAM OPERATOR NAME, ADDRESS, LOGO, AND WEBSITE	Program Operator Provided
GENERAL PROGRAM INSTRUCTIONS AND VERSION NUMBER	
LOCATION OF EXPLANATORY MATERIAL	
DECLARATION HOLDER AND ADDRESS	
DECLARATION NUMBER	Program Operator Provided
DECLARED PRODUCT & FUNCTIONAL UNIT	
PRODUCT DEFINITION	
REFERENCE PCR AND VERSION NUMBER	
MARKETS OF APPLICABILITY	
DATE OF ISSUE	Program Operator Provided
PERIOD OF VALIDITY	Program Operator Provided
EPD TYPE	[Product-specific]
EPD SCOPE	[Cradle to grave]
YEAR(S) OF REPORTED MANUFACTURER PRIMARY DATA	
LCA SOFTWARE & VERSION NUMBER	
LCI DATABASE(S) & VERSION NUMBER	
LCIA METHODOLOGY & VERSION NUMBER	
APPLICABLE GREEN BUILDING CERTIFICATION SCHEMA	
The PCR review was conducted by:	Program Operator Provided
	Program Operator Provided
	Program Operator Provided
This declaration was independently verified in accordance with ISO 14025: 2006. The UL Environment "Part A: Calculation Rules for the Life Cycle Assessment and Requirements on the Project Report," v3.0 (December 2017) , based on CEN Norm EN 15804 (2012) and ISO 21930:2017, serves as the core PCR, with additional considerations from the USGBC/UL Environment Part A Enhancement (2017)	
☐ INTERNAL ☐ EXTERNAL	Program Operator Provided
This life cycle assessment was conducted in accordance with ISO 14044 and the reference PCR by:	
	Program Operator Provided
This life cycle assessment was independently verified in accordance with ISO 14044 and the reference PCR by:	
	Program Operator Provided
LIMITATIONS Environmental declarations from different programs (ISO 14025) may not be comparable. Comparison of the environmental performance of products using EPD information shall be based on the product's use and impacts at the building level, and therefore EPDs may not be used for comparability purposes when not considering the building use phase as instructed under this PCR. Full conformance with the PCR for Water Chillers allows EPD comparability only when all stages of a life cycle have been considered, when they comply with all referenced standards, use the same sub-category PCR, and use equivalent scenarios with respect to construction works.. However, variations and deviations are possible". Example of variations: Different LCA software and background LCI datasets may lead to differences results for upstream or downstream of the life cycle stages declared.	

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Part B: Water Cooled Chillers



2. EPD Content

2.1. DESCRIPTION OF COMPANY/ORGANIZATION

The name of the manufacturing entity(ies) as well as the place(s) of production shall be provided. General information about the manufacturing entity(ies) may be provided, such as the existence of quality systems or environmental management systems, according to ISO 14001 or any other environmental management system in place.

2.2. PRODUCT DESCRIPTION

A narrative description of the product shall be provided that enables clear identification of the product. This description will include:

2.2.1 Product Identification

The declared products shall be identified by brand name(s), by material type(s), by production code(s) (if applicable), and by simple visual representation, which may be by photograph or graphic illustration.

2.2.2 Product Specification

Similar products grouped and reported as an average product in the same EPD satisfying the variation criteria of Part A, Section 5 shall constitute an individual declared product.

The appropriate ASTM or ANSI product specification shall be provided, including additional pertinent physical properties and technical information.

2.2.3 Flow Diagram

A graphical depiction of a flow diagram illustrating main production processes according to the scope of the declaration shall be included

2.2.4 Product Average

The method for creating a company specific individual product/product group EPD shall be described, including the method for determining a weighted average across products based on production volume. The manufacturing site(s) which the results of the LCA represent shall be specified.

Although chiller plant size may vary amongst installations, EPDs are product specific and the product family often covers a wide range of sizes. Because of this, a sampling from the product's potentially broad size range is required. Manufacturers should provide a representative sample size of their production from the previous calendar year for calculations, and should include up to 100 units to represent the range of sizes sold.

2.3. APPLICATION

The designated application(s) for the referenced product(s) shall be specified, along with the functional unit and Reference Service Life (RSL).

2.4. DECLARATION OF METHODOLOGICAL FRAMEWORK

The following items must be specified: the type of EPD with respect to life cycle stages, and the life cycle stages covered and not covered (i.e. cradle-to-grave with modules A1-C4 included).

The reference conditions for achieving the declared technical and functional performance and the RSL shall be included, per Part A: Life Cycle Assessment Calculation Rules and Report Requirements, Section 2.8.2.

The allocation procedures and cut-off procedure must be described. Include the statement "no known flows are deliberately excluded from this EPD."

2.5. TECHNICAL REQUIREMENTS

A listing of all standards required for the specification of the declared product shall be provided. A listing of all standards required for the testing, evaluation and approval of the declared product and its application in building



assemblies for building code and other regulation compliance shall also be provided and quoted as shown in Section 9.

The product(s) declared in this document complies with the following codes or regulations.

Example: AHRI Standard 550/590-2015, Performance Rating of Water-chilling and Heat Pump Water-heating Packages Using the Vapor Compression Cycle

The final evaluation report/certification/registration is available at: [insert link]

If relevant for the declared product, the following technical construction data for the product as delivered shall be provided with reference to the test standard.

TABLE 1. TECHNICAL DATA: WATER CHILLED COOLERS

Name	Value	Unit
Chilled Capacity		tons of refrigeration (RT) optional kW = 0.2843 * RT
Energy Efficiency		kW/ton at different combinations of load % and Entering Condenser Water Temperature (ECWT) as described in 'Energy Consumption' section below
Pressure Drops		ft of water optional kPa = 0.3346 * ft

2.6. PROPERTIES OF DECLARED PRODUCT AS DELIVERED

The dimensions/quantities of the declared product(s) as delivered to the site of installation/application shall be indicated.

2.7. MATERIAL COMPOSITION

The material composition of chiller products shall be disclosed and will include components as percentages or ranges of percentages of total mass as required by product Safety Data Sheet (SDS) rules, if relevant.

Statements of material non-inclusion, such as "... is free of ..." shall not be used.

Regulated Hazardous substances and dangerous substances shall be reported per Part A: Life Cycle Assessment Calculation Rules and Report Requirements, Section 4.11.

Note: This disclosure is intended to enable the user of the EPD to understand the composition of the product in delivery condition and support a safe and effective installation, use and disposal of the product. With appropriate justification, this requirement does not apply to confidential or proprietary information relating to materials and substances that apply due to a competitive business environment or covered by intellectual property rights or similar legal restrictions. It also might not be appropriate for information concerning intangible products.

2.8. MANUFACTURING

The manufacturing process and locations shall be described. If the EPD applies to several locations, the production processes for all locations shall be described and reference to quality management systems may be included.



2.9. ENVIRONMENT AND HEALTH DURING MANUFACTURING

Measures relating to environmental and health protection during the manufacturing process extending beyond national guidelines (of the production country) should be described, e.g. reference to a product safety data sheet (SDS), description of Environmental Management Systems or similar, programs addressing air emissions, waste water, noise, etc.

Example: During the entire production process, extra measures are taken to minimize the unintentional release of halogenated refrigerants. These measures are consistent with those identified in ANSI/ASHRAE Standard 147-2013, Reducing Release of Halogenated Refrigerants from Refrigeration and Air-Conditioning Equipment and Systems.

2.10. PACKAGING

Information on product-specific packaging: type, composition and possible reuse of packaging materials (paper, strapping, pallets, foils, drums, etc.) shall be included in this Section. The EPD shall describe specific packaging scenario assumptions, including disposition pathways for each packaging material by reuse, recycling, or landfill disposal based on packaging type.

In the absence of specific primary data, the data assumptions from Part A, Section 2.8.5, Table 2 shall be used.

In the case of reusable packaging designed to last for multiple reuse cycles, one reuse shall be assumed in the absence of primary manufacturer data. At the end of its reuse cycle, reusable packaging shall be assumed to go to landfill.

For non-reusable packaging, all plastic (e.g. film, scrap foam) and corrugated packaging at end-of-life is assumed to be disposed of in an appropriate landfill.

2.11. TRANSPORTATION

Specify type of transport; type of vehicle; distance; type and amount of energy carrier used to transport the product. Additional transportation elements are reported in Table 5.

Transport, installation, and deconstruction procedures are common to all products within the category. In the absence of primary data, the following assumptions should be used. Any deviations from these assumptions shall be justified and explained.

TABLE 2. TRANSPORT, INSTALLATION, AND DECONSTRUCTION PROCEDURES

Product transport from point of purchase to building site	Product transport from building site to waste processing	Installation & deconstruction procedures
Distance: 500 km Mode: Diesel-powered truck/trailer	Distance: 100 km Mode: Diesel-powered truck/trailer	Manual (no operational energy use)

2.12. PRODUCT INSTALLATION

A description of the type of processing, machinery, tools, dust extraction equipment, auxiliary materials, etc. to be used during installation and measures for reducing noise shall be included. Information on industrial and environmental protection may be included in this section.

Any waste treatment included within the system boundary of installation waste should be specified.

All quantitative information related to this section shall be reported in Section 4, "Scenarios and additional technical information."

2.13. USE

During the use phase, energy consumption of the chiller shall be calculated and documented according to the 'Centrifugal Chillers Average Efficiency Calculator', outlined in Section 3.3.1 "Energy Consumption".

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Any relevant information may be provided in this section regarding specific product use conditions and/or limitations relevant to each product application and/or use, including a description of the maintenance, repair, replacement or refurbishment process and/or a reference to where a description can be found.

All quantitative information related to this section shall be reported in Section 4, "Scenarios and additional technical information".

2.14. PRODUCT REFERENCE SERVICE LIFE AND BUILDING ESTIMATED SERVICE LIFE

The reference service life and building estimated service life shall be indicated according to Section 2.8.2 of Part A: Life Cycle Assessment Calculation Rules and Report Requirements.

The assumptions upon which the designated RSL is based and for which the RSL exclusively applies shall be provided in Section 4. Influences on aging, when applied, shall be in accordance with the state of the art.

The indication of the Reference Service Life (RSL) is imperative for EPDs covering the complete use stage (modules B1-B7), or if a use stage scenario is described, which refers to the lifetime of the product.

Based on industry convention, a default RSL of 25 years shall be assumed for water cooled chillers unless otherwise stated and substantiated (ASHRAE 2009).

When reporting the number of replacements necessary to fulfil the required performance and functionality over the building Estimated Service Life (ESL), fractional values shall be rounded up to the nearest tenth. As the RSL depends on in-use conditions according to ISO 21930, the EPD shall state such conditions along with the RSL or reference such conditions as described in other parts of the EPD when declaring the RSL. The product RSL shall not exceed the building ESL.

For product maintenance, reference shall be made to the manufacturer's maintenance manual for replacement of parts such as, but not limited to: starter and/or motor control (e.g. VFD). Refrigerant losses and replacement rates shall be assumed at 2% unless otherwise justified.²

2.15. RE-USE PHASE

The possibilities of re-use and recycling shall be described. If an Extended Producer Responsibility initiative such as a product take-back program exists, this may be included.

2.16. DISPOSAL

The possible disposal channels shall be indicated in accordance with disposal routes and waste classification referenced in Section 2.8.5 and 2.8.6 of Part A: Life Cycle Assessment Calculation Rules and Report Requirements.

3. LCA Calculation Rules

3.1. FUNCTIONAL UNIT

The functional unit for water cooler chillers is one (1) ton of chilling capacity with a provided conversion factor to 1 kg. The mass of one piece of the declared product shall be indicated (or the weighted average mass for group of products). The product sub-category (see Section II) shall be indicated. If accessories are required for installation of the product, their weighted-average mass must be included. If the accessories are made of different materials, it is permissible to group them by averaging together subject to requirements in Section 2.2.4.

² According to LEED v4 for Building Design and Construction, EA Credit: Enhanced Refrigerant Management. July 1, 2015.



TABLE 3. FUNCTIONAL UNIT

Name	Value	Unit
Functional unit	1 ton chilling capacity	
Mass of one delivered product		kg
Conversion factor to 1 kg chilling capacity		

3.2. SYSTEM BOUNDARY

The type of EPD shall be specified as cradle to grave, with all modules A1-A5, B1-B7, and C1-C4 reported. The modules considered in the Life Cycle Assessment as per “System limits” outlined in Section 6 of the PCR, Part A: Life Cycle Assessment Calculation Rules and Report Requirements shall be described in brief. Any relevant aspects or impacts not included in an information module shall be supported with relevant additional environmental information and the omissions shall be justified.

Capital goods and infrastructure flows shall not be excluded from unit processes used to model the LCIA to the extent they significantly affect the conclusions of the LCA or additional environmental information. The LCA report should specify lifetimes of capital goods and infrastructure included. The impact burden from capital goods and infrastructure shall be allocated to the product(s) in the LCA by either a) proportional to the specified lifetime of the asset, or b) proportional to the production output of the asset. Any deviation shall be explicitly specified and justified.

3.3. PRODUCT SPECIFIC CALCULATIONS FOR USE PHASE (MODULES B1-B7)

Use-stage environmental impacts of water chillers during building operations are significant and shall be determined according to the section below.

3.3.1. ENERGY CONSUMPTION

The total electric power consumption shall be calculated according to the Centrifugal Chiller Average Efficiency Calculator excel file. This file calculates the average efficiency of centrifugal chillers at AHRI conditions according to the method published in ASHRAE, 2009. Once efficiency is determined, the calculator is used to calculate the energy usage for each of 50 global city locations, which are independent of actual product markets. For positive displacement chillers, it is required to report chiller efficiency values at AHRI conditions as well as actual conditions. This is not the case with centrifugal chillers and a separate approach is used to adjust efficiencies.

Power consumption is calculated as follows in the Centrifugal Chiller Average Efficiency Calculator excel file:

Total electric power consumption (kWh) = Annual hours of operation (hrs/yr) × Efficiency (kW/ton) × Chiller capacity (tons) × Product life span (yrs)

To determine the annual hours of operation and efficiency, calculations in the Centrifugal Chiller Average Efficiency Calculator excel file are based on actual weather data from the following 50 global cities:

Atlanta, Bangkok, Beijing, Cairo, Cape Town, Caracas, Chicago, Dallas, Denver, Dubai, Hanoi, Ho Chi Minh, Hong Kong, Houston, Jerusalem, Kansas City, London, Los Angeles, Melbourne, Mexico City, Miami, Moscow, Mumbai, New Delhi, Ottawa, Paris, Perth, Phoenix, Riyadh, Rome, Sao Paulo, Seattle, Seoul, Shanghai, Singapore, Sydney, Taipei, Tokyo, Vancouver, Warsaw, and Washington, D.C.

The annual hours of operation and chiller capacity shall be defined as per the appended document ‘Centrifugal Chillers Average Efficiency Calculator.xlsx’.

Chiller capacity shall be defined as the chilling capacity declared on the product.

3.4. ESTIMATES AND ASSUMPTIONS

Key assumptions and estimates for interpretation of the Life Cycle Assessment should be referred to here, provided that they are not dealt with in Section 3 “LCA: Calculation rules”, or Section 4 “LCA: Scenarios and additional technical data”.

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3.5. UNITS

SI units are required for all LCA results. Other units commonly used in a regional market may be optionally included in addition to the required SI units.

TABLE 4. MANDATORY CONVERSION FACTORS TO BE USED IF OPTIONALLY REPORTING IN IMPERIAL UNITS

Convert from	To	Multiply by
square meter (m ²)	Square foot (ft ²)	10.76391
kilogram (kg)	Pound (lb)	2.204622
Mega joule (MJ)	British Thermal Unit (BTU)	947.8170
degree Celsius (°C)	degree Fahrenheit (°F)	$t/^{\circ}\text{C} = (t/^{\circ}\text{F} - 32)/1.8$
cubic meter (m ³)	cubic foot (ft ³)	35.31466

3.6. CUT-OFF CRITERIA

Cut-off rules as specified per the PCR, Part A: Life Cycle Assessment Calculation Rules and Report Requirements Section 2.9 shall be used and documented. All known mass and energy flows shall be reported. No known flows should be deliberately excluded.

3.7. DATA SOURCES

Data sources shall be documented per Part A: Life Cycle Assessment Calculation Rules and Report Requirements, Section 3.1.

3.8. DATA QUALITY

An evaluation shall be provided regarding data quality, including temporal, geographical, technological representativeness, and completeness and shall follow the requirements outlined in PCR, Part A: Life Cycle Assessment Calculation Rules and Report Requirements, Section 3.1.1.

3.9. PERIOD UNDER REVIEW

The period under review and ensuing averages shall be documented.

3.10. ALLOCATION

Part A, Section 3.3 shall be used as the basis for allocation decisions, and mass should be used as the primary basis for co-product allocation in this Part B. Allocation methods deemed more appropriate than on the basis of mass may be used but only when justified. The allocations of relevance for calculation (appropriation of impacts across various products) shall be indicated, at least:

- Allocation in the use of recycled and/or secondary raw materials
- Allocation of energy, ancillary and operating materials used for individual products in a factory

whereby reference shall be made to the modules in which the allocations are performed.

3.11. COMPARABILITY AND BENCHMARKING

Comparison of EPD results between non-competitive products may be included this section per the requirements in Part A: Life Cycle Assessment Calculation Rules and Report Requirements, Section 9.

Systems with large differences in capacity should not be compared.

4. LCA: Scenarios and additional technical information

The following information is necessary for the declared modules. Irrelevant or non-applicable module rows may be excluded in the EPD; additional information may also be listed if necessary

The following technical information is a basis for the declared modules or can be used for developing specific scenarios in the context of a building assessment if modules are not declared (MND).

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TABLE 5. TRANSPORT TO THE BUILDING SITE (A4)

Name	Value	Unit
Liters of fuel		l/100km
Transport distance		km
Capacity utilization (including empty runs)		%
Gross density of products transported or weight and volume of products transported		kg/m ³
Capacity utilization volume factor		-

TABLE 6. INSTALLATION INTO THE BUILDING (A5)

Name	Value	Unit
Auxiliary materials		kg
Net freshwater consumption specified by water source and fate (amount evaporated, amount disposed to sewer)		m ³
Other resources		kg
Electricity consumption		kWh
Other energy carriers by type		MJ
Product loss per functional unit		kg
Waste materials at the construction site before waste processing, generated by product installation		kg
Output materials resulting from on-site waste processing (specified by route; e.g. for recycling, energy recovery and/or disposal)		kg
Direct emissions to ambient air, soil and water		kg

* Method used to calculate and report airborne dust shall be identified and/or described

TABLE 7. USE OR APPLICATION OF THE INSTALLED PRODUCT (B1)

Name	Value	Unit
RSL		years

TABLE 8. MAINTENANCE (B2)

Name	Value	Unit
Maintenance process information (cite source in report)		-
Maintenance cycle		Number/ RSL
Maintenance cycle		Number/ ESL
Water consumption specified by water source and fate (amount evaporated, amount disposed to sewer)		m ³
Auxiliary materials specified by type (e.g. cleaning agent)		kg
Other resources		kg

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Electricity consumption		kWh
Other energy carriers specified by type		MJ
Refrigerant replacement, specified by type		kg
Material loss		kg
Direct emissions to ambient air, soil and water		kg
Further assumptions for scenario development (e.g. frequency and time period of use, number of occupants);		

TABLE 9. REPAIR (B3)

Name	Value	Unit
Repair process information (cite source in report)		-
Inspection process information (cite source in report)		-
Repair cycle ³		Number/ RSL
Repair cycle		Number/ ESL
Water consumption specified by water source and fate (amount evaporated, amount disposed to sewer)		m ³
Auxiliary materials specified by type (e.g. cleaning agent)		kg
Other resources		kg
Electricity consumption		kWh
Other energy carriers specified by type		MJ
Material loss		kg
Direct emissions to ambient air, soil and water		kg
Further assumptions for scenario development (e.g. frequency and time period of use, number of occupants);		As appropriate

4.1. REPLACEMENT (B4) / REFURBISHMENT (B5)

The number of replacements of product expected during the building ESL of 75 years shall be declared by dividing the building ESL by the product RSL and subtracting one. Required or expected maintenance is to be modelled according to manufacturer's guidelines. Assumptions and key parameters shall be clearly stated and the manufacturer is to submit supporting documentation to justify the assumptions made.

Using the default assumptions for product RSL and building ESL provided in this PCR, two replacement cycles are calculated. The corresponding impacts reported in module B4 will be approximately twice the total score of modules A1-A5, assuming two replacements are made.

TABLE 10. REPLACEMENT (B4)

Name	Value	Unit
Replacement cycle(s) ⁴		ESL/RSL -1
Electricity consumption		kWh
Other energy carriers specified by type		MJ

³ Repair, Replacement, and Refurbishment cycles should be rounded-up to the nearest tenths; e.g., 1.47 rounded to 1.5.

⁴ Ibid.



Water consumption specified by water source and fate (amount evaporated, amount disposed to sewer)		m ³
Replacement of worn parts		kg
Direct emissions to ambient air, soil and water		kg
Further assumptions for scenario development, e.g. frequency and time period of use, # occupants		As appropriate

TABLE 11. REFURBISHMENT (B5)

Name	Value	Unit
Refurbishment process description (cite source in report)		
Refurbishment cycle ⁵		Number/ RSL
Refurbishment cycle ⁶		Number/ ESL
Electricity consumption		kWh
Other energy carriers specified by type		MJ
Water consumption specified by water source and fate (amount evaporated, amount disposed to sewer)		m ³
Material inputs during refurbishment		kg
Waste material		kg
Direct emissions to ambient air, soil and water		kg
Further assumptions for scenario development, e.g. frequency and time period of use, # occupants		As appropriate

TABLE 12. OPERATIONAL ENERGY USE (B6) AND OPERATIONAL WATER USE (B7)

Name	Value	Unit
Net freshwater consumption specified by water source and fate (amount evaporated, amount disposed to sewer)		m ³
Electricity consumption		kWh
Other energy carriers		MJ
Equipment output		kW

TABLE 13. END OF LIFE (C1-C4)

Name	Value	Unit
Collected separately		kg
Collected as mixed construction waste		kg
Reuse		kg
Recycling		kg
Energy recovery		kg
Landfilling		kg

⁵ Ibid.

⁶ Ibid.



TABLE 14. REUSE, RECOVERY AND/OR RECYCLING POTENTIALS , RELEVANT SCENARIO INFORMATION

Name	Value	Unit
Net energy benefit from energy recovery from waste treatment declared as exported energy in C3 (R>0.6)		MJ
Net energy benefit from thermal energy due to treatment of waste declared as exported energy in C4 (R<0.6)		MJ
Net energy benefit from material flow declared in C3 for energy recovery		MJ

5. LCA: Results

In Table 15, **Error! Reference source not found.** "Description of the system boundary," all declared modules shall be indicated with an "X".

Modules A1, A2, and A3 may be declared as one aggregated module A1-A3.

Per Part A, life cycle impact assessment (LCIA) results shall be declared using scientific notation with three significant digits (e.g. $1.23E-5 = 0.0000123$) for each module. Uniform formatting shall be used for all indicator values.

- ▶ North America (Part A, Section 4.7, Table 7, TRACI indicators)
- ▶ EU (Part A, Section 4.8, Table 8, CML indicators)
- ▶ Rest of World (Part A, Section 4.9, Table 8, indicators as provided)

Results derived from the product life cycle inventory (LCI) shall be reported as follows:

- ▶ Resource use indicators (Part A, Section 4.1, Table 4)
- ▶ Output flows and waste category indicators (Part A, Section 4.1.2, Table 5)
- ▶ Carbon emissions and removals (Part A, Section 4.6, Table 6)

TABLE 15. DESCRIPTION OF THE SYSTEM BOUNDARY MODULES

	PRODUCT STAGE			CONSTRUCTION STAGE		USE STAGE							END OF LIFE STAGE				BENEFITS AND LOADS BEYOND SYSTEM BOUNDARY
	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
	Raw material supply	Transport	Manufacturing	Transport from gate to site	Assembly/Install	Use	Maintenance	Repair	Replacement	Refurbishment	Operational Energy Use During Product Building	Operational Water Use During Product Use	Deconstruction	Transport	Waste processing	Disposal	Reuse, Recovery, Recycling Potential
Cradle to grave	Required					Required (based on scenarios)							Required				Not allowed

6. LCA: Interpretation

Interpretation requirements for the Project Report are provided in Part A: Life Cycle Assessment Calculation Rules and Report Requirements, Section 5.

An interpretation shall be provided in the EPD which discusses the assumptions and limitations associated with the interpretation of results as declared in the EPD, both methodology and data related.

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This interpretation shall also include a description of the time frame and/or variance of the LCIA results if the EPD is valid for several products. An illustration of the results with figures is recommended, e.g. for the dominance analysis, the distribution of impacts across the modules, the CO₂-balance, etc. as appropriate for a reader's understanding of the environmental profile of the declared product.

7. Additional Environmental Information

7.1. ENVIRONMENT AND HEALTH DURING MANUFACTURING

Measures relating to environmental and health protection during the product manufacturing process extending beyond national guidelines (of the production country) may be described, e.g. reference to a product safety data sheet (SDS), description of Environmental Management Systems or similar, programs addressing air emissions, waste water, noise, etc.

7.2. ENVIRONMENT AND HEALTH DURING INSTALLATION

Information should be provided in this section on the relationship between the product, the environment and health, including any possible harmful substances or emissions e.g. reference to a product safety data sheet (SDS). Any recommendations concerning cleaning, maintenance, etc. of the declared product should be listed in Section 4 "Technical information on scenarios".

7.3. EXTRAORDINARY EFFECTS

FIRE

Information should be included on the product's fire performance and possible impacts on the environment e.g. reaction-to-fire, other relevant fire tests as applicable, and emissions to air.

WATER

Information should be included on the product's performance and possible impacts on the environment following unforeseeable influence of water, e.g. flooding.

MECHANICAL DESTRUCTION

Information should be included on the product's performance and possible impacts on the environment following unforeseeable mechanical destruction.

7.4. DELAYED EMISSIONS

If a manufacturer wishes to declare quantitative or qualitative information on delayed emissions used to calculate Global Warming Potential within the EPD, information may be provided here. See Part A, Section 4.4 for more information.

7.5. ENVIRONMENTAL ACTIVITIES AND CERTIFICATIONS

Other environmental activities, such as participation in recycling or recovery programs along with the details of these programs and contact information, shall be provided.

For certifications applied to the product and listed in the EPD, a statement shall be included on where an interested party can find details of the certification program.

7.6 FURTHER INFORMATION

A reference source for additional information may be provided here, e.g. homepage, reference source for safety data sheet.

8. Supporting Documentation

The Project Report Content, Structure, and Accessibility requirements to support an EPD created using this document are provided in Part A: Section 2. Project Report elements include general information (Part A: Section 2.1), study goal (Part A: Section 2.2), study scope (Part A: Section 2.8), and the life cycle inventory analysis, impact assessment, and interpretation (Part A: Section 3, 4, and 5). Additionally, the Project Report shall include additional required supporting documentation specified in this sub-category Part B and according to Part A: Section 6.

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If relevant to the scope of the declared product, or due to the product material composition, it is recommended to provide sufficient supporting documentation in the EPD and Project Report. When providing documentation, testing protocols and other relevant information shall be indicated. If supporting documentation is not provided, the reasons shall be indicated in the EPD and Project Report.

All statements shall be documented with measured data (presented by the corresponding test certificates). In the case of non-verifiable substances, the limit of detection shall be included in the declaration. Interpreting statements such as "... free of ..." or "... are entirely harmless ..." are not permissible.

9. References

The literature referred to in the Environmental Product Declaration shall be quoted in full from the following sources. Standards and standards relating to evidence and/or technical features already fully quoted in the EPD do not need to be listed here. Part B of the PCR document on which they are based shall be referenced.

UL ENVIRONMENT

UL Environment General Program Instructions April 2017, version 2.1

Part A: Life Cycle Assessment Calculation Rules and Report Requirements UL Environment (February 2018, version 3.1)

CHARACTERIZATION METHODS

IPCC. 2014. CLIMATE CHANGE 2013. THE PHYSICAL SCIENCE BASIS. CAMBRIDGE UNIVERSITY PRESS.
[HTTP://WWW.IPCC.CH/REPORT/AR5/WG1/](http://www.ipcc.ch/report/ar5/wg1/)

Hauschild M.Z., & Wenzel H. Environmental Assessment of Products. Springer, US, Vol. 2, 1998

Heijungs R., Guinée J.B., Huppes G., Lankreijer R.M., Udo de Haes H.A., Wegener Sleeswijk A. Environmental Life Cycle Assessment of Products: Guide and Backgrounds. CML. Leiden University, Leiden, 1992

Jenkin M.E., & Hayman G.D. Photochemical ozone creation potentials for oxygenated volatile organic compounds: sensitivity to variations in kinetic and mechanistic parameters. Atmospheric Environment. 1999, 33 (8) pp. 1275–1293

WMO. 1999. Scientific Assessment of Ozone Depletion: 1998, World Meteorological Organization Global Ozone Research and Monitoring Project – Report No. 44, WMO, Geneva

SUSTAINABILITY REPORTING STANDARDS

EN 15804: 2012-04 - Sustainability of construction works — Environmental Product Declarations — Core rules for the product category of construction product.

ISO 14025: 2006 - Environmental labels and declarations — Type III environmental declarations — Principles and procedures

ISO 14040: 2006 - Environmental management – Life cycle assessment – Principles and framework

ISO 14044:2006 - Environmental management – Life cycle assessment – Requirements and guidelines

ISO 14046:2013 - Environmental management- Water footprint- Principles, requirements and guidelines

ISO 15392:2008 - Sustainability in building construction- General principles

ISO 15686-1:2011 - Buildings and constructed assets- Service life planning- Part 1: General principles

ISO 15686-2:2008 - Buildings and constructed assets- Service life planning Part 2: Service life prediction procedures

ISO 15686-7:2008 - Buildings and constructed assets- Service life planning Part 7: Performance evaluation for feedback of service life data from practice

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ISO 15686-8:2008 - Buildings and constructed assets- Service life planning Part 8: Reference service life and service life estimation

ISO 21930: 2017 - Sustainability in building construction -- Environmental declaration of building products

LEED v4 for Building Design and Construction, EA Credit: Enhanced Refrigerant Management. July 1, 2015.

Product Category Rule Guidance Development Initiative. Guidance for Product Category Rule Development. (August 28, 2014, version 1.0).

TESTING AND CLASSIFICATION REFERENCES

ASHRAE 140 -- Standard Method of Test for the Evaluation of Building Energy Analysis Computer Programs

AHRI 550/590 -- Performance Rating of Water-chilling and Heat Pump Water-heating Packages Using the Vapor Compression Cycle

ASHRAE Journal. Geister, R. and Thompson, M. A Closer Look at Chiller Ratings. December 2009.

RELEVANT FEDERAL STANDARDS AND SOPS

Environment Canada, National Pollutant Release Inventory (NPRI) (<http://www.ec.gc.ca/inrp-npri/>)

EPCRA 313 Toxic Release Inventory Reporting (U.S.) (<http://www2.epa.gov/toxics-release-inventory-tri-program>)

US EPA, ORD/NRMRL/Sustainable Technology Division, Systems Analysis Branch, SOP No. S-10637-OP-1-0- Tool for the Reduction and Assessment of Chemical and other Environmental Impacts (TRACI), Software Name and Version Number: TRACI version 2.1, USER'S MANUAL, 24 July, 2012

US: Resource Conservation and Recovery Act (RCRA), Clause C (<http://www.epa.gov/region6/rcra/>)

40 CFR 50 Protection of Environment - Part 50: National Primary and Secondary Ambient Air Quality Standards (U.S.) (<http://www.epa.gov/air/criteria.html>)

Clean Air Act (CAA) Section 112(r): Accidental Release Prevention/Public Management Rule (http://www.epa.gov/oem/docs/chem/caa112_rmp_factsheet.pdf)

CERCLA Hazardous Substances (U.S.) (<http://www.epa.gov/oem/content/hazsubs/cerclsubs.htm>)

U.S. Department of Labor, Occupational Safety & Health Administration (OSHA 1910.1200 Hazard Communication Standard—Toxic and Hazardous Substances (U.S) (http://www.osha.gov/pls/oshaweb/owadisp.show_document?p_table=standards&p_id=10099)

RELEVANT PCRS

Part A: Life Cycle Assessment Calculation Rules and Report Requirements UL Environment (February 2018, version 3.1)

Product Category Rules for Water Chilled Coolers. Institute of Construction and Environment e.V., Königswinter (May 2011, version 1.0)